

CE 552: Concrete Technology

Course Code	CE 552
Course Name	Concrete Technology
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Distribution	Discipline Elective
Approval	9th Senate (May 2016)

Course Contents

- **History of Concrete construction** [1 Lecture]
- **Introduction to Special concretes:**
 - High strength/performance concrete: Principle, Ingredient properties, Material selection, Design principles, Production, Curing, Properties in fresh state and hardened states, Durability.
 - Self-consolidating concrete: Benefits and Limitations, Properties in fresh and hardened states, design principles and Quality Control.
 - Fibre reinforced concrete: Types of fibres, Structure of matrix, Fibre-Cement interactions, Cracking mechanics.
 - Light weight concrete: Production and properties of No-fines concrete, Aerated and foamed concrete, Lightweight aggregate concrete.
 - Recycled aggregate concrete: Use of industrial, construction and demolition wastes and their effect on concrete properties. [6 Lectures]
- **Cement:** Manufacture, Types, Aspects of hydration. [5 Lectures]
- **Supplementary cementing materials:** Fly ash, Silica fume, Ground granulated blast furnace slag, Metakaolin, Rice Husk Ash ' Characteristics, Properties of mortar and concrete in fresh and hardened states, Durability of concrete. [5 Lectures]
- **Chemical admixtures:** Water reducing agents, Super-plasticizers, Air entrainers, Accelerators, Damp proofers and Corrosion inhibitors. [4 Lectures]
- **Aggregates** -Types and Properties. [4 Lectures]
- **Principles of mix design.** [3 Lectures]
- **Concrete production, Placement and Curing:** Equipment and practices; Formwork for concrete structures: Economy, Pressures on formwork, Forms for beams, columns, slabs and footings, Slipforms and Failures of formwork. [5 Lectures]
- **Module IX:** Introduction to the testing of fresh, mechanical and durability characteristics of concrete. [5 Lectures]
- **Module X:** Introduction to Prescriptive and Performance specifications. [2 Lectures]
- **Quality control:** Nature of variability, Control charts. [2 Lectures]

Text Books

1. A. M. Neville, **Properties of concrete**, 4th Edition, Pearson Education, 1963.
2. Neville and Brooks, **Concrete technology**, Prentice Hall, 2006.
3. Mehta, P.K. and Monteiro, P.J.M., **Concrete: structure, properties and materials**, Prentice Hall, NJ, 2006.
4. Mindess, S., Young, J.F. and Darwin, D., **Concrete**, Prentice Hall, NJ, 2003.

References

1. Siddique, P. and Khan, M.I., **Supplementary cementitious materials**, Springer, 2011.
2. Rixom, R. and Mailvaganam, N., **Chemical admixtures for concrete**, E & FN SPON, 2002.
3. Aitcin, P.C., **High performance concrete**, E & FN SPON, 2011.
4. Bentur, A. and Mindess, S., **Fibre reinforced cementitious composites**, E & FN SPON, 2006.
5. Brito, J.de and Saikia, N., **Recycled aggregate in concrete: use of industrial, construction and demolition waste**, Springer, 2012.
6. Day, K.W., **Concrete mix design, quality control and specification**, E & FN SPON, 2006.
7. Peurifoy, R. L. and Oberlender, G. D., **Formwork for concrete structures**, McGraw Hill, 1996.
8. Dackzo, J.A., **Self-consolidating concrete**, Spon press, 2012.